Table 1: Example fully observable NeuronBench task (z_rebound). The target is the membrane vector field $\dot{V}$, regressed from the external current I_{ext}, voltage V, and channel open fractions $\phi_{\text{Na}} = m_{\text{Na}}^3 h_{\text{Na}}$, $\phi_{\text{K}} = n_{\text{K}}^4$, and $\phi_{\text{Z}} = m_{\text{Z}}^2 h_{\text{Z}}$. Every task shares the Hodgkin–Huxley sodium, potassium, and leak currents and adds one channel; here, a low-threshold inward current Z that is de-inactivated by hyperpolarization.

$$\dot{V} = \underbrace{I_{\text{ext}} - 120\,\phi_{\text{Na}}(V - 50) - 36\,\phi_{\text{K}}(V + 77) - 0.3\,(V + 54.4)}_{\text{shared Hodgkin–Huxley currents}} \underbrace{- 4\,\phi_{\text{Z}}(V - 120)}_{\text{new channel}}$$